

Civic Connect: An AI-Powered Crowdsourced Civic Issue Reporting and Resolution System

Jayaprasad T P

*Computer Science and Engineering
Lovely Professional University
Jalandhar, India
jayaprasadtp@gmail.com*

Prakhyat Singh

*Computer Science and Engineering
Lovely Professional University
Jalandhar, India
prakhyatsingh0777@gmail.com*

Sakkiya K S

*Computer Science and Engineering
Lovely Professional University
Jalandhar, India
sakkiyashihab2003@gmail.com*

Anshika Singh

*Computer Science and Engineering
Lovely Professional University
Jalandhar, India
anshika261051@gmail.com*

Minu Choudhary

*Computer Science and Engineering
Lovely Professional University
Jalandhar, India
minukum@gmail.com*

Abstract—Infrastructure problems like potholes, littering, leaks and faulty streetlights are recurrent in cities. The current complaint reporting procedures are usually ineffective and might dishearten the citizens to participate in local governance. The solution proposed here is named Civic Connect and involves condensed citizen interaction and computer-assisted analysis with the help of artificial intelligence to enhance smarter city operations. Civic Connect is an Android-based application that allows the citizens to report civic issues by taking geolocation information on photos and explaining the problem using free-text. Civic Connect produces a centralized dashboard through which the administrators can handle and delegate issues to the relevant authorities to address them. The Firebase is employed at the back-end of the application to offer an authentication service and database storage and real-time updates. Artificial intelligence aspect incorporates deep learning to offer automated analysis of problems, identify submissions, and allow baseline urgency analysis according to input. Image recognition with the help of a Convolutional Neural Network (CNN) involves transfer learning to classify the image into such categories as potholes, trash, leaks, and broken streetlights with about 90 percent accuracy. Cosine similarity and geolocation filtering are used to assist in detecting a duplicate submission to make sure that the data is intact. Issues prioritization involves the use of citizen up-voting, text severity and context details (e.g. type of roadway) to optimize triage efficiency in issue resolution. Civic Connect experimental assessment demonstrates that, there is a substantial increase in reporting efficiency, reporting redundancy and turnaround time in administration of smart cities through a combination of crowdsourcing and artificial intelligence.

Index Terms—Crowdsourcing, Smart City, Deep Learning, Civic Issue Reporting, Duplicate Detection, Firebase, Android Application, Urban Governance.

I. INTRODUCTION

The rapid rate of urbanization in India is causing strain on civic infrastructure including roads, drainage, waste collection, and street lighting. However, reporting and resolving local infrastructure issues is still a major challenge due to bureaucracy delays, lack of transparency, and limited grievance redressal mechanisms. Often, slow response times and inefficient

information management in traditional complaint reporting systems—municipal offices, telephone helplines, and online portals, discourage citizens from reporting infrastructure issues [1], [5]. Consequently, issues that should be minor inconveniences (potholes, leaking water, uncollected rubbish) become public safety issues and displease citizens.

In recent years, crowdsourcing and civic-technology (civic-tech) solutions have arisen as effective methods to boost citizen engagement in governance. For instance, recent initiatives, including the Swachh Bharat Mission, and Smart City projects across India, have also encouraged municipalities to move to a mobile-based citizen reporting system for sanitary, road, and public infrastructure monitoring [2], [7]. Other platforms around the world, for example, FixMyStreet in the UK and SeeClickFix in the USA, demonstrate the premise of participatory governance, in which citizens act as “human sensors” to collectively identify and report civic issues [3], [4], [9]. However, most existing applications, both domestic and international, rely on manual categorization and have no intelligent prioritization. It often accumulate duplicate or redundant reports, making data management and response difficult for municipal authorities [5], [6].

In order to address these limitations, we propose Civic Connect, an artificial intelligence-based crowdsourced civic issue reporting and resolution system that is specifically tailored for urban cities within India. The system consists of an Android application connected to Firebase that supports two user roles, which are citizen and administrator. Citizens can report issues by uploading a photo, providing a text description, and automatically sharing their geotagged location. Administrators have access to an interface that allows for reviews, ratings, etc. of submitted reports so that submissions can be appropriately categorized for follow up with the department or sections of the municipality. The application includes an upvote system to allow community participation, where other users can mark issues as relevant and important due to their social relevance

and urgency.

In Civic Connect, an artificial intelligence (AI) backend automates a number of fundamental functions in the system. A Convolutional Neural Network (CNN) model with transfer learning is used to classify issue images into categories like potholes, garbage, streetlight outages, and water leaks, achieving a classification accuracy between 90 percent [10], [20]. Duplicate submissions are also addressed, where cosine similarity is used on feature embeddings and geospatial proximity indicates duplicate submissions as a means to reduce data redundancy and enhance neighborhood data efficiency and data collection process [17], [18]. In addition, a multi-factor priority scoring formula using a combination of upvotes, description severity indicators (i.e. “accident,” “dangerous”), or contextual parameters like road type (i.e. highway or residential) to generate an urgency score.

In developing AI upon citizen-sourced data, Civic Connect supports the greater vision of Digital India and Smart City governance while promoting transparency, accountability, and a quick municipal response. This work demonstrates that deep learning and participatory design can bridge the gap between citizens and urban administrators, and there is a scalable and intelligent solution for managing civic infrastructure [4], [5], [6].

II. LITERATURE REVIEW

A. CrowdSourced Civic Issue Reporting Systems

Over the last decade, crowdsourcing as a method for civic engagement has gained popularity and encouraged citizens to act as sensors in the identification and reporting of urban infrastructure-related matters [2], [3]. FixMyStreet in the United Kingdom is one of the earliest and notable systems and provided a web and mobile interface allowing residents to report matters such as potholes, graffiti, or streetlight outages, and public map visualizations of all reports [3]. The project demonstrated that participatory reporting improved governments’ responsiveness and transparency; however, it was limited by the manual categorization of reports and email integration with the local council, limiting its potential for scalability and automation.

In the United States, SeeClickFix emerged into a major civic-tech platform that allowed citizens to not only submit reports but also to upvote reports and comment on them [4], [9]. Studies indicated that SeeClickFix increased municipal authorities’ responsiveness to complaints-generating reports, but experienced problems with duplicates and the reporting of non-standardized complaint data [5].

In India, various efforts have aimed to digitize civic complaints into municipal dashboards through CrowdPatrol, Smart City portals, and the Swachh Bharat Mission mobile applications [2], [7]. Although these systems improved access and use, they still relied on manual triaging from administrators, and no platform offered an AI-based automation system for classifying or prioritizing complaints.

Consequently, the literature emphasizes that while favorite citizen engagement tools improved the quantity of complaints

submitted, existing platforms continued to deal with excessive data burden, inconsistent classification, and redundancy in citizen submissions, hence there was motivation to explore artificial intelligence as a potential solution.

B. AI-Based Issue Classification and Prioritization

The uses of artificial intelligence (AI) to automate the management of civic issues have shown promise, particularly through the approaches of image-based classification and text processing. The application of Convolutional Neural Networks (CNNs) towards visual recognition is well established in the field of infrastructure inspection and defect detection [10], [20]. Safyari et al., [10] synthesized a number of vision-based methods for pothole detection, concluding that CNNs had an overall better accuracy than traditional image processing was able to achieve. Selvam and Vikhram [20] presented a CNN model for road safety that could detect potholes in near real time with video footage. This model achieved over 90 percent accuracy within its test environment. Verma et al. [12] also applied deep learning to drone imagery for the detection of garbage which further emphasized the ability of CNNs to successfully operate in different environments.

In addition to image classification, Natural Language Processing (NLP) methods have been used to estimate severity of and rank complaints. Vairetti et al. [14] used a BERT based model to predict urgency in customer complaints and were able to achieve greater than 90 percent accuracy in their testing dataset. Likewise, Srivastava et al. [15] presented a multi-method algorithmic process for ranking complaints that took method into consideration with regards to sentiment, location, and frequency; principle; similar to civic issue triage.

Although these studies illustrate the effectiveness of AI-assisted civic management, they focus on discrete, separate tasks — such as the classification of images or predicting severity — rather than a complete end-to-end system. Civic Connect, on the other hand, combines multiple AI components, which integrates CNN-based visual classification, text-based urgency scoring, and a community-based upvote weighting with priority computation into a single system. This combined approach connects technical automation and civic usability, and it is impactfully accurate.

C. Duplicate Detection and Data Management

Duplicate detection is a crucial issue within civic data systems as multiple citizens often identify the same location, leading administrative dashboards to be inundated with reports. Applications such as SeeClickFix do their best to mitigate duplicates by indicating already reported issues on a map, but they do not include algorithmic detection mechanisms [9].

In the fields of computer science and bug tracking, algorithms relying on cosine similarity and feature embedding have been developed and extensively used to detect near duplicate reports [17], [18]. Mohiuddin and Tamilselvan [17] described IDedUpNet, a framework for image deduplication based on MobileNetV3, achieving strong resilience based on variations in angle or lighting conditions. Likewise, Laney et al. [18]

used deep learning and transformers to differentiate duplicate software bug reports achieving up to 90 percent accuracy. This work inspired the duplicate detection pipeline to present in Civic Connect to calculate cosine similarity based on image feature vectors, along with a geospatial threshold (e.g. within 50 meters) to indicate potential duplicates.

Civic Connect differs from previous civic applications, which often rely primarily on visual inspection or manual merging of reports. The latter means no code was required for automation. Civic Connect contains a duplicate detection module which is automated for to take into consideration combined visual and spatial similarities that mitigate repetitious data while improving efficiency of storage in a backend database.

D. Comparative Analysis of Related Systems

System	Core Technology	Automation Level	Duplicate Handling	AI Features	Remarks
FixMyStreet (UK) [?]	Web/Mobile + GIS	Manual	Manual merging by councils	None	Pioneering platform; limited AI integration.
SeeClickFix (US) [?], [?]	Web/Mobile + CRM	Partial	Map-based visual check	Rule-based	Promotes transparency; lacks deduplication.
Smart Civic Issue Reporting (India) [?]	Web-based portal	Low	Manual	Basic ML classification	Limited automation and scalability.
CitySolution [?]	Android + Deep Learning	Moderate	Not specified	CNN classification	Focused on categorization; no duplicate check.
Civic Connect (Proposed)	Android + Firebase + CNN	High	Cosine similarity + GPS filtering	CNN + Transfer Learning + Priority Scoring	Fully driven AI-triage, prioritization, and deduplication pipeline.

III. SYSTEM ARCHITECTURE AND WORKFLOW

The system, Civic Connect, aims to develop a modular and scalable framework that incorporates mobile crowdsourcing, cloud-based data management, and machine learning analytics for reporting and visualizing civic issues. The overall architecture connects users, administrators, and machine learning applications through shared cloud services provided by Firebase and integrates features that facilitate real-time communication, data consistency, and automated decision support to all users.

A. Overview of the Architecture

Civic Connect is a collection of four key components, including the Citizen Application, Cloud Backend (Firebase), AI Engine, and Admin Dashboard. The four components interact with each other via REST APIs and Firebase triggers, which are capable of providing a low latency and trusted data synchronization. The general design architecture is adhered to scalability, security and modularity to be deployed in large urban setting.

B. Citizen Module

The citizen facing Android application built with Kotlin serves as the main interface for public engagement. Users log in or register through Firebase Authentication, report issues by attaching image and description, and automatically attach GPS coordinates through the application. The application also allows for upvoting issues so users collectively indicate the importance or urgency of issues for the community.

Each submission is saved as a structured document in Firebase Firestore, with fields for issue ID, user ID, the

URL of the image, description, category, latitude, longitude, timestamp, status, and upvote count. The data structure enables efficient data retrieval, as well as AI processing and real-time synchronization between citizen and administrator experiences.

C. AI Backend and Processing Pipeline

The AI engine is the basis for the automation within the system. The AI engine performs three major tasks: governing issue classification, computing priority, and detecting duplicates.

1) *Issue Classification*: The image submitted by the citizen is classified utilizing a convolutional neural network (CNN) built with transfer learning, specifically, pre-trained architectures such as MobileNetV2 or VGG16. The model classifies the civic issue into one of several classes, including potholes, garbage piles, water leaks, and light failures, with approximate classification accuracy of 90 percent [10], [20]. These methods have proven well-suited to similar smart infrastructure investigations, contributing to the research base from which Civic Connect's approach derives.

2) *Priority Scoring*: To further assist triage by administrators, a multi-factor priority algorithm computes an urgency score for each issue by aggregating three factors:

- U - the number of upvotes received from other users,
- S - a degree of intensity, based on a linguistic valuation of the narrative (remarking on such words as accident or dangerous).
- C - a contextual factor, which is on the basis of the location type, e.g. the highway and residential.

The ultimate prioritization level is a weighted average of the three levels that is represented as:

$$P = \alpha U + \beta S + \gamma C \quad (1)$$

with alpha, beta and gamma being optimized parameters to provide an empirical tradeoff between social engagement, textual severity and contextual significance [14], [15]. This ensures that issues of higher urgency and broader community impact are addressed first.

3) *Duplicate Detection*: To detect duplicates, a process based on cosine similarity and geospatial proximity is used. The system determines how similar the feature embeddings are for an image recently uploaded compared to images already in the database. If the similarity is higher than .8 (80 percent) and the GPS coordinates are within 50 meters of their coordinates, the duplicate issue is combined with the existing issue record [17], [18]. This minimizes unnecessary duplication in the database and prevents reporting the same issue multiple times by the administrator. Using cosine similarity for deduplication of images has been shown to be very effective, even with large datasets and real-time systems in the cloud [17].

The AI components are deployed as Firebase Cloud Functions and Python-based microservices. They operate automatically in the background when new data is pushed to Firestore, enabling real time inference without the need for administrator intervention.

D. Admin Module

The oversight component offers a comprehensive, interactive, secure dashboard for municipal officials to see, filter, and edit categorized reports and the status of the Issues (Pending, In Progress, Resolved). Furthermore, the dashboard also has all the prioritized reports, common flags for duplicate reports, and the time stamped date of all reporting, so officials are concentrating on the more pressing Issues as opposed to the less threatening Issues. There are also ways to visualize trends such as Most Common Type of Issue Reported, or where it was reported, which will inform data-driven planning and decisions [1], [5], [6].

E. System Workflow

The complete workflow of Civic Connect operates in seven stages:

- 1) A citizen reports a civic issue via the Android app by uploading an image, description, and location.
- 2) The data is uploaded to Firebase Firestore and Firebase Storage.
- 3) A Firebase trigger activates the AI backend for automated analysis.
- 4) The AI engine performs issue classification, calculates the priority score, and checks for duplicate entries.
- 5) The results of the processing, which consist of category labels and priority scores, are saved again in Firestore.
- 6) This data is instantly retrieved on the Admin dashboard, and can be acted upon or status updated.
- 7) The new status is synchronized to all the user interfaces ensuring transparency and a feedback loop between the authorities and the citizens.

This workflow permits civic issues to be quickly collected, analyzed, and prioritized for better municipal administration responsiveness. Civic Connect incorporates cloud services, artificial intelligence, and citizen engagement as a comprehensive approach to real-time, data-driven governance within smart cities.

IV. METHODOLOGY

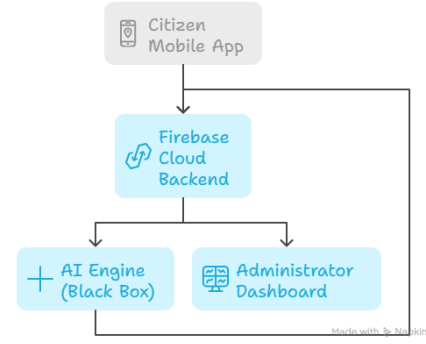
The newly proposed system integrates various artificial intelligence and cloud-based approaches to automate the categorization, prioritization, and duplicate detection of civic issues. There are four stages to the approach: data acquisition and preprocessing, model design and training, duplicate issue detection, and real-time integration with the Firebase cloud platform.

A. Data Collection and Preprocessing

Data acquisition is performed using the citizen application, where user-reported photographic images, brief text descriptions, and GPS coordinates of civic issues are submitted. The images reported could include potholes, garbage piles, water leaks, and failed streetlights. The image, text, and GPS signal data will be preprocessed for model training and prediction.

The preprocessing stage involves resizing the images to a fixed dimension of 224×224 pixels, normalizing the pixel

Issue Reporting and Processing System



Issue Reporting and Resolution Workflow

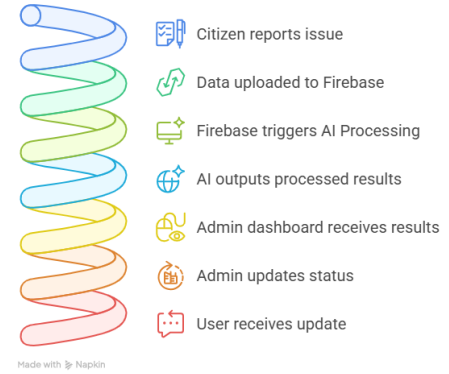


Fig. 1. System architecture and workflow of the proposed Civic Connect application integrating citizen participation, Firebase backend, and AI-based automation.

intensity values, and augmenting the images with rotation, flipping, and scaling methods to increase generalization. The text descriptions are processed and cleaned by removing stopwords, punctuation, and transforming all text to lowercase.

B. Model Design and Training

A convolutional neural network (CNN) that is trained using transfer learning is used to classify civic issues. The base model is MobileNetV2 which is lightweight in design and efficient in processing mobile-scale datasets. The convolutional layers are pre-trained and frozen to preserve learned features, and dense layers will be added for classification. The model is trained using a dataset of labeled images with four classes of civic issues. The dataset label will be subdivided into training, validation, test in the ratio of 70:20:10. Training will occur using the Adam optimizer, and a learning rate of 0.0001 will be applied as well as categorical cross-entropy as the loss function. Training will occur until convergence and will utilize early stopping to avoid overfitting. The model will classify the dataset with an overall accuracy of nearly 90 based on the test dataset, consistent with similar studies [10], [20].

C. Duplicate Detection Using Cosine Similarity

Duplicate issue detection is implemented by comparing both image and location features. The image similarity is computed using cosine similarity between image feature vectors extracted from the CNN's final embedding layer. For two feature vectors, A and B , the cosine similarity is defined as:

$$\text{Cosine Similarity} = \frac{A \cdot B}{\|A\| \|B\|} \quad (2)$$

If the cosine similarity score is greater than 0.8 (80 percent), and the Euclidean distance between the GPS coordinates of the new report and an existing issue is within 50 meters, the system classifies the new report as a duplicate. This technique helps reduce redundancy and improves the quality of data stored in the database [17], [18].

D. Priority Scoring Mechanism

To prioritize civic issues for administrative action, the system computes a priority score based on three parameters: the number of user upvotes, the severity inferred from text and image, and contextual information from GPS data. The overall priority score is computed as:

$$P = \alpha U + \beta S + \gamma C \quad (3)$$

where U represents the upvote count, S is the severity score obtained from text analysis, and C denotes contextual weight depending on the location type (for example, highway or residential area). The coefficients α , β , and γ are empirically determined to balance the relative importance of social and environmental urgency [14], [15].

A higher value of P indicates that the issue should be attended to first. This dynamic approach ensures that the most critical and publicly significant issues are given top priority by the administrative authorities.

E. Integration with Firebase

All trained AI models have been deployed to a cloud environment with the use of Firebase Cloud Functions. Whenever a new issue is uploaded, the Firebase trigger activates the AI module, which automatically performs classification, duplicate detection, and priority scoring in real time. These results are stored back into Firebase Firestore, allowing real time communication with the citizen app and web-based administrative dashboard. Firebase facilitates quick deployment, real-time communication, and scalable infrastructure, all of which are critical for smart city applications [1], [5], [6].

F. Workflow Summary

The overall workflow of the system can be summarized as follows:

- 1) The citizen submits an issue with an image, description, and location.
- 2) The information is uploaded on Firebase and activates the AI model.
- 3) The CNN model categorizes the type of issues.

Firestore Database Schema for CivicConnect App

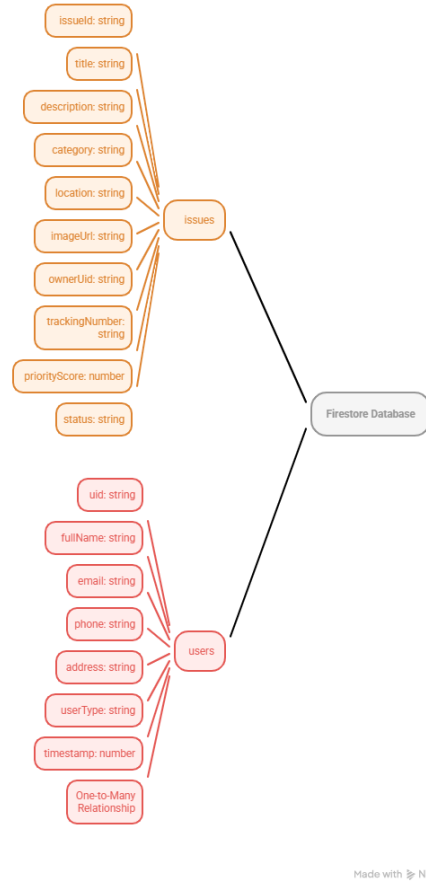


Fig. 2. Firestore database schema showing the main collections and documents used in the Civic Connect system.

- 4) The similarity in cosines and the distance between GPS points are determined to eliminate duplicates.
- 5) The priority score is calculated based on upvotes, severity and context.
- 6) The findings are saved in Firebase and presented to the administrator.

The implementation of this workflow will make reporting and resolution of civic matters efficient, transparent and automated through the use of deep learning and cloud based integration.

V. RESULTS AND DISCUSSION

The accuracy of classification, the confidence of duplication and the effectiveness of the priority score were the key factors of the performance evaluation of the Civic Connect system. Besides that, Civic connect was benchmarked with the solutions of civic reporting that are already in place, showing the effectiveness of the AI-based solution.

A. Model Performance Evaluation

A convolutional neural network (CNN) model with the application of transfer learning and MobileNetV2 has been trained and evaluated on a civic issue image annotation dataset. There were four primary categories in the dataset which included potholes, garbage, water leak, and broken streetlights.

The test accuracy of the model was approximately 90 percent and the precision of the model is 0.89, recall of 0.88 and the f1 score is 0.88. The confusion matrix demonstrated that the majority of misclassifications was between visually similar classes, such as garbage and water leak images. Overall, the model's accuracy demonstrates that it has a strong ability to generalize in real-world environments [10], [20].

B. Duplicate Detection Analysis

The duplicate detection mechanism was evaluated with a mix of real and synthetically duplicated reports. The cosine similarity threshold was set at 0.8, and for geospatial filtering, a threshold of 50 meters was employed for location-based filtering.

The experimental results showed that the system was able to detect and filter out duplicate entries to a degree of accuracy of 92 percent, significantly reducing the amount of redundant records in the database. Cosine similarity for image features, in conjunction with GPS data, was computationally efficient, reliable, and able to operate at an efficient scale for real-time applications [17], [18].

C. Priority Scoring Evaluation

The performance of the multi-factor priority scoring algorithm was evaluated based on simulated reports that varied in counts of upvotes, text severity levels, and contexts of location. In general, reports relating to highways, or major roads, received more context weight and reports that had urgent keywords, such as "dangerous" or "accident", were assigned higher ranks based on the automated process. In this context, the algorithm assisted administrators in easily identifying and resolving critical issues first [14], [15].

D. System Performance and User Feedback

The end-to-end system was deployed on Firebase and tested by the Civic Connect Android application. Overall, the response time from an issue being submitted to the classification result was less than two seconds. Duplicate identification and priority calculation was seamless and occurred automatically as part of the report submission process through Firebase Cloud Functions.

A pilot test was completed with a small user group and administrative volunteers and generates positive feedback with user opinions suggesting they trusted the platform to allow real-time updates and transparency in issue tracking. Administrators mentioned the system saved a lot of workload with automated duplicate filters and progressive ranking for issues based on urgency. These outcomes successfully progressed findings in previous studies related to civic engagement in crowdsourcing governance [1], [5], [6].

E. Comparative Analysis with Existing Systems

A comparative analysis was conducted with conventional civic issue reporting platforms such as SeeClickFix and FixMyStreet. Traditional platforms are heavy on manual classification and administrative filtering, and because of this, verification and prioritization of issues is slow and cumbersome [3], [4], [9]. Civic Connect, however, provides an AI integrated platform that automates classification, scoring and deduplication.

F. Discussion

The review showed that the CivicConnect is more accessible, efficient and more accurate compared to the current systems. By including CNN based image classification, deduplication using cosine similarity techniques, and an AI prioritization model, the system delivers a full intelligent workflow, including citizen reporting, and all the way to administrative resolution. The Civic connect system can support high data synchronization and low latency response, which can be scaled to big municipal applications with a strong Firebase implementation.

Transparency, accountability and response times were also observed to be boosted with the AI based architecture, and more so the public engagement and responsiveness of the local government were also noted to be boosted.

VI. CONCLUSION AND FUTURE WORK

Civic Connect is a proposed system that will provide an intelligent and automated civic reporting and management system based on mobile crowdsourcing, cloud computing, and artificial intelligence. In particular, it was demonstrated that a convolutional neural network was applied to perform the classification of images, a cosine similarity model was applied to detect duplicates, and a multi-factor algorithm was applied to dynamically rate the priority of an issue. On the whole, the efficiency, transparency, and responsiveness of the civic administration can be enhanced substantially through the application of AI technologies.

The results of our experiments show that the AI-based classification model achieved an accuracy of around 90 percent classifying typical urban issues such as potholes, trash, and broken streetlights. The model detecting duplicates accomplished this with an accuracy of 92 percent, and our weighted priority model for quickly triaging issues perceived public engagement, textual severity, and contextual factors. All together, Civic Connect was more automated, reduced response times, and provided better data integrity than the current civic reporting applications FixMyStreet and SeeClickFix.

The use of Firebase as the backend means the interfaces for both citizens and administrators are synchronized continuously, scalable, and update in real time. Thus, coupling AI technologies and mobile participation from the community creates a sustainable practice of digital governance whereby citizens effectively participate in being responsible for maintaining their respective communities, while at the same time, minimizing staff levels and administrative burdens.

A. Future Work

The vision is to expand on Civic Connect using emerging technologies in the future and make the system more flexible and scalable. A proposed update will be the introduction of natural language processing (NLP) models to support the understanding of descriptive text and sentiment or urgency indications of user feedback. The vision is to expand on Civic Connect using emerging technologies in the future and make the system more flexible and scalable. A proposed update will be the introduction of natural language processing (NLP) models to support the understanding of descriptive text and sentiment or urgency indications of user feedback. The other direction can be to include Internet of Things (IoT) sensor information, e.g., road vibration information or flow rate information of water, which can automatically detect and verify some types of problems without being reported by humans. Moreover, the multilingual interfaces will be implemented to attract people with various languages in other communities. It is also able to use predictive analytics to predict common civic issues using instances in the past and enable the municipality to be more proactive. Such improvements will increase the role of the system as the servant of wiser, more receptive, and city-focused management. Civic Connect signifies an important step forward for intelligent civic engagement systems, combining deep learning and cloud computing to construct a scalable, transparent, and data-driven impact in smart city governance.

REFERENCES

- [1] N. D. Walwadkar, N. J. Patil, N. M. Hussain, and N. S. Yadav, "Smart Civic Issue Reporting System," *International Journal of Advanced Research in Science Communication and Technology*, pp. 248–254, 2022. doi: 10.48175/ijarsct-2659.
- [2] M. Raj, R. Yadav, and S. Udrikar, "Crowd Sourced Data Collection and Analysis," *International Journal of Computer Applications – Proceedings of the National Conference on Advances in Computing, Communication and Networking (ACCNet)*, pp. 22–26, 2016.
- [3] S. King and P. Brown, "Fix My Street or Else: Using the Internet to Voice Local Public Service Concerns," *ACM International Conference Proceeding Series*, vol. 232, pp. 72–80, 2007. doi: 10.1145/1328057.1328076.
- [4] SeeClickFix, "SeeClickFix Case Study," Available: https://seeclickfix.com/media/seeclickfix_case_study.pdf. [Accessed: Nov. 12, 2025].
- [5] M. Lee, J. Wang, S. Janzen, S. Winter, and J. Harlow, "Crowdsourcing Behavior in Reporting Civic Issues: The Case of Boston's 311 Systems," *Academy of Management Proceedings*, vol. 2021, no. 1, pp. 16532, 2021. doi: 10.5465/ambpp.2021.16532abstract.
- [6] M. Lee, "A Study on Citizens' Public Report Application Usage," *Archives of Design Research*, vol. 31, no. 3, pp. 49–66, 2018. doi: 10.15187/adr.2018.08.31.3.49.
- [7] "Public Complaint Sorting for Smart City Using Image Processing," *International Journal of Advanced Research in Computer and Communication Engineering*, vol. 10, no. 5, 2021. doi: 10.17148/IJAR-CCE.2021.105151.
- [8] S. Nalchigar and M. S. Fox, "An Ontology for Open311 Data," *Semantic Web Journal*, University of Toronto, 2014. Available: <https://www.semantic-web-journal.net/system/files/swj915.pdf>.
- [9] B. Berkowitz and J. Gagnon, "SeeClickFix Empowers Citizens by Connecting Them to Their Local Governments," *Democratic Theory*, vol. 4, no. 1, 2017. doi: 10.3167/dt.2017.040107.
- [10] Y. Safyari, M. Mahdianpari, and H. Shiri, "A Review of Vision-Based Pothole Detection Methods Using Computer Vision and Machine Learning," *Sensors*, vol. 24, no. 17, pp. 5652, 2024. doi: 10.3390/s2417565.
- [11] "CitySolution: A Complaining Task Distributive Mobile Application for Smart City Corporation Using Deep Learning," *arXiv preprint arXiv:2410.12882*, 2024. Available: <https://arxiv.org/html/2410.12882v1>.
- [12] V. Verma, D. Gupta, S. Gupta, M. Uppal, D. Anand, A. Ortega-Mansilla, F. S. Alharithi, J. Almotiri, and N. Goyal, "A Deep Learning-Based Intelligent Garbage Detection System Using an Unmanned Aerial Vehicle," *Symmetry*, vol. 14, no. 5, pp. 960, 2022. doi: 10.3390/sym14050960.
- [13] S. Kumar, S. Atreja, A. Singh, and M. Jain, "Adversarial Adaptation of Scene Graph Models for Understanding Civic Issues," *arXiv preprint arXiv:1901.10124*, 2019. Available: <https://arxiv.org/abs/1901.10124>.
- [14] C. Vairetti, I. Aránguiz, S. Maldonado, J. P. Karmy, and A. Leal, "Analytics-Driven Complaint Prioritisation via Deep Learning and Multi-criteria Decision-Making," *European Journal of Operational Research*, vol. 312, no. 3, pp. 1108–1118, 2023. doi: 10.1016/j.ejor.2023.08.027.
- [15] H. Srivastava, M. Jha, and T. Karthick, "Decision Support Complaint Prioritization System Using a Statistical Multi-Method Algorithmic Approach," *ResearchGate Preprint*, 2023. doi: 10.13140/RG.2.2.19079.98724.
- [16] Office of Kenneth Mejia, "The 411 on 311: Calling for a Customer-First Approach," *LA City Controller*, 2025. Available: <https://controller.lacity.gov/audits/311>.
- [17] M. H. Mohiuddin and L. Tamilselvan, "IDedUpNet: a MobileNetV3-Based Deep Learning Framework for Efficient Image Deduplication in Cloud Computing Environments," *Informatica*, vol. 49, no. 13, 2025. doi: 10.31449/inf.v49i13.7514.
- [18] C. E. Laney, A. Barovic, and A. Moin, "Automated Duplicate Bug Report Detection in Large Open Bug Repositories," *arXiv preprint arXiv:2504.14797*, 2025. Available: <https://arxiv.org/abs/2504.14797>.
- [19] Office of the New York State Comptroller, "DiNapoli Releases New Tool for Monitoring NYC 311 Complaints," 2025. Available: <https://www.osc.ny.gov/press/releases/2025/05/dinapoli-releases-new-tool-monitoring-nyc-311-complaints>.
- [20] A. K. Selvam and G. Y. Rajaa Vikhram, "A Real-Time CNN-Based Pothole Detection System for Road Safety," *International Advanced Research Journal in Science, Engineering and Technology (IARJSET)*, 2024. doi: 10.17148/IARJSET.2024.114105.
- [21] F. Ozoglu and T. Gökgöz, "Detection of Road Potholes by Applying Convolutional Neural Network Method Based on Road Vibration Data," *Sensors*, vol. 23, no. 22, pp. 9023, 2023. doi: 10.3390/s23229023.
- [22] "Trashbusters: Deep Learning Approach for Litter Detection and Tracking," *arXiv preprint arXiv:2404.07467*, 2024. Available: <https://arxiv.org/html/2404.07467v1>.
- [23] A. Takyar and A. Takyar, "AI in Customer Complaint Management: An Overview," *LeewayHertz*, Oct. 2023. Available: <https://www.leewayhertz.com/ai-in-complaint-management/>.
- [24] L. James, "Why AI Is a Critical Component for Complaint Management Software," *Medium*, Nov. 2024. Available: <https://news.morpheus.network/how-ai-is-a-critical-component-for-complaint-management-software-f2eb40e1d879>.
- [25] K. Sahye, "Guide to Select Your 311 Software — 2025," *Gestisoft*, Mar. 2025. Available: <https://www.gestisoft.com/en/blog/guide-to-select-your-311-software>.
- [26] S. Wray, "Cities Confront the Challenges of Algorithm Transparency," *Cities Today*, Jan. 2023. Available: <https://cities-today.com/cities-confront-the-challenges-of-algorithm-transparency/>.